SOMAIYA VIDYAVIHAR UNIVERSITY

02·05·2025 (E)

Semester: January 2025 – April 2025		
Maximum Marks: 50 **Examination: End-Semester Examination**		**Duration: 2 Hrs.**
Programme code: 41 **Programme: B.Tech** CCE	**Class: SY**	**Semester: IV** (SVU-2023)
Institute/School/Department: K. J. Somaiya School of Engineering	**Name of the department/Section/Center: Computer and Communication Engg (CCE)**	
Course Code: 216U41C402	**Name of the Course: RF Fundamentals and Applications**	
Instructions: 1) Draw neat diagrams 2) All questions are compulsory 3) Assume suitable data wherever necessary		

Que. No.	Question Statement	Max. Marks
Q.1	Attempt any **two**	
i)	A transmission line has a reflection coefficient Γ=0.3 and characteristics impedance 50Ω. Calculate: a) Return Loss (in dB) b) Voltage Standing Wave Ratio (VSWR) c) Load impedance d) load admittance	05
ii)	List various types of transmission lines used in RF communication. Explain two wire transmission line model with primary constants.	05
iii)	Explain five different types of antennas with their applications in RF domain.	05
Q.2	Attempt any **one**	10
i)	Explain high frequency behaviour of capacitor with the help of circuit diagram, equations and frequency response. Also discuss chip capacitors.	
ii)	Design a maximally flat low pass filter with cut of frequency 2 GHz, impedance of 50Ω and at least 15dB insertion loss at 3GHz. (Refer graph and table on Page 2)	
Q.3	Attempt any **one**	10
i)	State and explain FRISS transmission equation. A transmitting antenna radiates a power of 50 mW at a frequency of 2.4 GHz. The gain of the transmitting antenna is 5 dB, and the gain of the receiving antenna is 3 dB. The distance between the antennas is 10 meters. Calculate the received power using the Friis transmission equation. (Assume free-space conditions)	
ii)	Explain following characteristics of antenna: a) Polarization b) Radiation intensity c) Effective area of aperture d) Radiation pattern	

Q.4	Attempt the following	10
	a) Explain the importance of RF measurement and testing in communication systems. Mention any three commonly used RF testing instruments. b) Explain the terms with their significance in RF: Noise figure and noise factor	
Q.5	Attempt the following	10
	a) Explain various types of propagation losses. b) Discuss with example the effect of microwaves on living and non-living things.	

Maximally Flat Filter Tables & Attenuation Response

Table 5-2 Coefficients for maximally flat low-pass filter ($N = 1$ to 10)

N	g_1	g_2	g_3	g_4	g_5	g_6	g_7	g_8	g_9	g_{10}	g_{11}
1	2.0000	1.0000									
2	1.4142	1.4142	1.0000								
3	1.0000	2.0000	1.0000	1.0000							
4	0.7654	1.8478	1.8478	0.7654	1.0000						
5	0.6180	1.6180	2.0000	1.6180	0.6180	1.0000					
6	0.5176	1.4142	1.9318	1.9318	1.4142	0.5176	1.0000				
7	0.4450	1.2470	1.8019	2.0000	1.8019	1.2470	0.4450	1.0000			
8	0.3902	1.1111	1.6629	1.9615	1.9615	1.6629	1.1111	0.3902	1.0000		
9	0.3473	1.0000	1.5321	1.8794	2.0000	1.8794	1.5321	1.0000	0.3473	1.0000	
10	0.3129	0.9080	1.4142	1.7820	1.9754	1.9754	1.7820	1.4142	0.9080	0.3129	1.0000

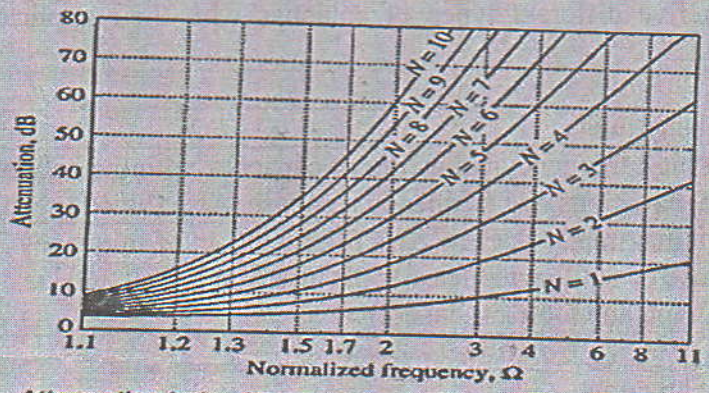

Figure 5-18 Attenuation behavior of maximally flat low-pass filter versus normalized frequency.